

HACKATHON PROJECT SUBMISSION

PAK-ANGELS MID PROGRAM HACKATHON

PROJECT TITLE

FreshSense

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4 MEMBERS

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Product Requirements Document (PRD) • September 2026

FreshSense

Deep Learning & Computer Vision Platform for Automated Food Freshness Diagnostics

A unified engineering and product specification for an AI-powered visual food freshness classification engine. FreshSense leverages deep convolutional neural network embeddings (MobileNetV2) trained over agglomeratively clustered multi-source datasets to categorize produce across five discrete quality degradation stages in under one second.

PROJECT STATUS	TARGET AUDIENCE	MODEL ARCHITECTURE	DATE & VERSION
MVP Deployed (Active)	Consumers, Grocers, Agritech	MobileNetV2 (Agglomerative)	September 2026 • v2.1.0

01 Executive Summary & Vision

Food spoilage and wastage represent a multi-billion dollar economic and environmental challenge worldwide. Approximately one-third of all food produced for human consumption is lost or wasted annually, with produce (fruits and vegetables) exhibiting the highest wastage rates. A major contributor is the absence of accessible, objective, and automated tools for consumers and retailers to accurately evaluate the freshness lifecycle before irreversible spoilage occurs.

FreshSense bridges this gap by deploying modern computer vision directly to web and edge interfaces. Users upload or capture produce imagery, which is preprocessed and evaluated by an optimized MobileNetV2 deep learning classifier. Rather than relying on simple binary (Fresh/Rotten) heuristics, FreshSense scores the item across **five biological degradation stages** (Fresh, Slightly Aged, Stale, Spoiled, Rotten) with confidence scoring and persistent audit history.

82%

MACRO F1 SCORE

13

FOOD SPECIES

5

FRESHNESS STAGES

< 850ms

INFERENCE LATENCY

02 Problem Statement & Value Proposition

2.1 The Core Problem

- Subjective Human Inspection:** Traditional freshness evaluation relies on manual sensory estimation (touch, smell, surface sight), which varies significantly across individuals and conditions.
- Binary Simplification:** Existing consumer-facing classification systems reduce food quality to binary Fresh vs. Rotten states, failing to inform users when produce is in early degradation ("Slightly Aged" or "Stale") where salvage or prompt consumption is viable.
- Cloud Latency & Privacy:** Many ML offerings require massive GPU cloud clusters, adding latency, data privacy concerns, and heavy hosting infrastructure costs.

2.2 Proposed Solution & Value Proposition

FreshSense provides a lightweight, client-accessible inference service powered by an 11.5 MB optimized MobileNetV2 model running locally over CPU utilizing Keras 3 with NumPy/JAX execution. The system features:

- 1. **Multi-Class Granularity:** Five distinct stages reflecting real culinary and microbiological states.
- 2. **Explainable Probability Distribution:** Full visibility into confidence metrics across all five classes.
- 3. **Local & Resilient Execution:** Zero GPU requirement, zero external API dependencies, and high execution stability across operating systems.
- 4. **Audit Logging:** Chronological scan archive allowing quality tracking across batches and dates.

03 Target Personas & Key User Journeys

USER PERSONA	PAIN POINTS & GOALS	PRIMARY FRESHSENSE WORKFLOW
Home Consumer <i>(Priya, Meal Planner)</i>	Unsure whether produce in refrigerator has degraded or remains safe to feed family. Aims to minimize grocery budget waste.	Takes a smartphone photo of suspicious produce → drops into web interface → observes "Slightly Aged (89%)" → decides to cook into soup immediately rather than discard.
Retail Produce Lead <i>(Marco, Grocery Manager)</i>	Daily shelf sorting requires fast decisions on markdowns, food bank donations, or composting. Needs standardized consistency.	Scans crate sample images → flags items scoring "Stale" for immediate 50% discount shelf → routes "Spoiled" lots for immediate vendor credit claim.
Agritech Supply Auditor <i>(David, Logistics QC)</i>	Receiving bulk cold-storage shipments; requires fast spot audits to accept/reject distributor lots with timestamped proof.	Uploads multiple batch inspection images → reviews confidence logs → exports audit history table as objective inspection documentation.

04 Functional Requirements (FR)

4.1 Image Ingestion & Input Pipeline

- **FR-1.1 (Dropzone Ingestion):** The UI shall provide an interactive Drag-and-Drop area with visual drag-over feedback and click-to-browse fallback.
- **FR-1.2 (Format Validation):** The system must accept standard web formats: .jpg, .jpeg, .png, and .webp. Unsupported formats must return a clean client error without crashing.
- **FR-1.3 (Client Preview & Dismissal):** Upon file selection, an instant client-side thumbnail preview shall render with a dedicated dismiss button to cancel or re-upload.

4.2 Computer Vision Preprocessing

- **FR-2.1 (Standardized Geometry):** All uploaded imagery must be programmatically converted to RGB and resized to exact dimensions of $128 \times 128 \times 3$ pixels.
- **FR-2.2 (Floating-Point Normalization):** Pixel values must be cast to 32-bit floats and scaled between $[0.0, 1.0]$ via division by 255.0.
- **FR-2.3 (Batch Dimension Packaging):** Input tensors must be formatted as (1, 128, 128, 3) matching Keras model tensor requirements.

4.3 Deep Learning Quality Inference Engine

- **FR-3.1 (Multi-Class Prediction):** The engine must evaluate input tensors and output a Softmax probability vector across five classes: Fresh, Slightly Aged, Stale, Spoiled, and Rotten.
- **FR-3.2 (Argmax Class Assignment):** The designated prediction class shall correspond to the highest probability score.
- **FR-3.3 (Silent Inference Mode):** Inference calls must execute with `verbose=0` to eliminate stdout terminal escape sequences and prevent Windows C runtime I/O exceptions (Errno 22).

4.4 Explainable Results & Data Presentation

- **FR-4.1 (Primary Status Badge):** Results must display a prominent, color-coded status badge with associated percentage confidence score.
- **FR-4.2 (Actionable Culinary Guidance):** Each category must render an automated plain-language recommendation regarding safety and consumption.
- **FR-4.3 (Confidence Distribution Bars):** All five classes must render individual animated horizontal progress bars showing the model's computed probability distribution.

4.5 Diagnostic History & Persistent Audit Log

- **FR-5.1 (Persistent Archival):** Each successful scan must be stored in `history.json` with a unique UUID, original filename, static image URI, prediction class, confidence score, full probability dictionary, and formatted timestamp.
- **FR-5.2 (History Invalidation & Cleanup):** The system must provide a `DELETE /history` endpoint that purges records and flushes uploaded temporary artifacts.

05 Technical Architecture & Data Flow

```

+-----+ +-----+ +-----+
| Client Browser | POST | Flask Server | | MobileNetV2 AI Engine |
| (HTML5 / JS / CSS) | ----> | (Python 3.14) | ----> | (Keras 3 NumPy Backend) |
+-----+ +-----+ +-----+

| | | |
| 1. Drag & Drop Image | 2. Save image to static folder | 4. Execute Inference |
| (JPG / PNG / WebP) | 3. Resize 128x128 & Normalize | (Silent verbose=0) |
| | | |
| | | <-----+ |
| | | 5. Softmax Class Probabilities |
| | | 6. Persist to history.json |
| <-----+ |
| 7. JSON Response: { class, confidence, probabilities, timestamp } |
| 8. Animate distribution bars & render color-coded badge |

```

Subsystem Layer	Technology Component	Architectural Justification & Role
Deep Learning Core	MobileNetV2 (Agglomerative)	11.5 MB model weight footprint; depthwise separable convs; high F1-score (0.82) with low CPU overhead.
Inference Engine	Keras 3.15 + NumPy Backend	Zero TensorFlow / CUDA dependency; runs on standard hardware environments with immediate startup.
Application Server	Python 3.14 + Flask 3.1	Lightweight WSGI/REST server with minimal latency overhead for multipart file streaming and JSON APIs.
Image Processing	Pillow (PIL 12.3)	Fast C-optimized image decoding, conversion, and bicubic interpolation at 128x128.
Frontend Client	Semantic HTML5 + Custom CSS + ES6	Dependency-free single page application with modern typography, glassmorphism-free clean aesthetic.

06 Machine Learning Pipeline & Benchmark Evaluation

6.1 Curated Multi-Source Dataset

The underlying training corpus incorporates a merged and normalized collection of three public Kaggle food freshness repositories (AlineSellwia, Swoyam2609, and Muhridin Muxiddinov), totaling **6.41 GB** across 13 primary produce categories:

- **Fruits:** Fresh & Rotten variants of Apple, Banana, Mango, Orange, Strawberry.
- **Vegetables:** Fresh & Rotten variants of Bell Pepper, Bitter Gourd, Capsicum, Carrot, Cucumber, Okra, Potato, Tomato.
- **Standard Transformations:** Noise reduction using Gaussian blur, Canny edge detection, brightness adjustments, horizontal flipping, and $\pm 15^\circ$ rotations to prevent overfitting.

6.2 Clustering Validation: Agglomerative vs. K-Means

Because natural food decay is a continuous biological spectrum rather than a hard boundary, unsupervised clustering was employed on image feature embeddings to discover consistent degradation stages. Two primary strategies were trained across 8 model architectures:

MODEL ARCHITECTURE	K-MEANS ACCURACY	K-MEANS MACRO F1	AGGLOMERATIVE ACCURACY	AGGLOMERATIVE MACRO F1
AlexNet	0.72	0.70	0.84	0.83
MobileNetV2 (Selected)	0.77	0.75	0.82	0.82
DenseNet121	0.75	0.74	0.83	0.82
Xception	0.72	0.70	0.81	0.80
InceptionV3	0.74	0.72	0.74	0.73
VGG16	0.66	0.64	0.70	0.57
VGG19	0.68	0.65	0.66	0.53
ResNet50	0.37	0.23	0.42	0.34

Model Selection Rationale: MobileNetV2

While AlexNet demonstrated marginally higher raw benchmark accuracy (0.84 vs 0.82), the pre-trained weights in the original repository were stored as Git LFS pointer stubs. MobileNetV2 provided a fully verified, intact 11.5 MB binary with identical macro F1 precision (0.82) and 10× smaller parameter complexity, making it the ideal production selection for real-time web deployment.

07 Freshness Taxonomy & Diagnostic Criteria

CATEGORY	BADGE	BIOLOGICAL MARKERS	CULINARY & SAFETY RECOMMENDATION
Fresh	Fresh	Intact cellular firmness, uniform pigmentation, zero superficial pitting or mold.	Prime consumption condition. Safe for raw intake or prolonged storage.
Slightly Aged	Slightly Aged	Minor water loss, mild skin dullness, minor softening at extremities.	Safe for raw or cooked consumption. Consume within 24–48 hours.
Stale	Stale	Pronounced dehydration, noticeable loss of aroma, wrinkled surface.	Culinary quality degraded. Best cooked, stewed, or pureed; not for raw salad.
Spoiled	Spoiled	Tissue softening, early microbial colonies, off-color patches.	Unfit for human consumption. Risk of foodborne illness; discard immediately.
Rotten	Rotten	Advanced fungal mycotoxins, structural collapse, pungent decomposition.	Hazardous bio-waste. Discard immediately into compost or sealed waste.

08 Non-Functional Requirements (NFR)

8.1 Performance & Latency

- **NFR-1.1 (Latency Budget):** Total request lifecycle (upload + preprocess + inference + history write + response) shall complete in $\leq 1,200$ ms on standard quad-core CPU hardware.
- **NFR-1.2 (Cold Start Time):** Model loading during application boot shall not exceed 3.5 seconds.

8.2 Platform Reliability & I/O Isolation

- **NFR-2.1 (Silent Keras Execution):** The inference loop must avoid console terminal formatting calls to prevent Windows CRT EINTR (Errno 22).
- **NFR-2.2 (Resource Leaks):** All opened PIL image descriptors and file stream pointers must be enclosed in context managers (with `Image.open(...)`) to guarantee instant handle release on Windows file systems.
- **NFR-2.3 (Encoding Consistency):** All JSON history read and write operations must be explicitly configured with `encoding="utf-8"`.

8.3 Accessibility & Design System Compliance

- **NFR-3.1 (Aesthetic Standard):** Strict avoidance of generic templates, artificial emoji UI icons, or illegible glassmorphism transparencies. Adherence to a clean Figma-level aesthetic featuring high contrast, custom SVG iconography, and harmonious typography.
- **NFR-3.2 (Color Contrast):** All typography must satisfy WCAG 2.1 AA minimum contrast ratios ($\geq 4.5:1$ for standard text, $\geq 3:1$ for large display headings).
- **NFR-3.3 (Cross-Device Responsiveness):** Full responsive fluidity from 375px mobile viewports up to 1440px desktop displays.

09 Risk Analysis & Mitigation Matrix

IDENTIFIED RISK	SEVERITY / LIKELIHOOD	MITIGATION STRATEGY IMPLEMENTED
Windows I/O Errno 22	High / High	Enforced <code>verbose=0</code> on <code>model.predict()</code> , suppressing Keras Progbar cursor manipulation and box-drawing character outputs to non-interactive handles.
Non-Standard Produce Angles	Medium / Medium	Training dataset incorporated comprehensive spatial augmentations ($\pm 15^\circ$ rotation, 180° flip, brightness variability) to maximize angle invariance.
File Lock on History Write	Medium / Low	Wrapped history file operations in try/except blocks with explicit UTF-8 encoding so minor audit log I/O glitches never prevent diagnostic result delivery.
Unsupported File Formats	Medium / Low	Multi-tiered validation checking MIME type on client side, file extensions on server side, and Pillow decoding verification with graceful error handling.

10 Product Roadmap & Future Milestones

RELEASE PHASE	TIMELINE	TARGET CAPABILITIES & DELIVERABLES
Phase 1 (MVP)	Current Release	MobileNetV2 engine, 5-stage classification, drag-and-drop web UI, confidence charts, persistent JSON history log, and responsive design.
Phase 2 (Streaming)	Q4 2026	Live WebRTC camera feed integration for real-time continuous video classification without manual photo capture. Multi-item bounding box detection.
Phase 3 (Edge Mobile)	Q1 2027	Export model to TFLite (TensorFlow Lite) and ONNX formats for offline on-device mobile applications (iOS / Android) with zero server latency.
Phase 4 (Enterprise)	Q3 2027	Commercial grocery store dashboard, barcode/SKU association, shelf-life forecasting time-series estimation, and smart refrigerator IoT integration.

11 Document Approval & Governance

STAKEHOLDER ROLE	NAME / ORGANIZATION	SIGN-OFF STATUS	DATE
Lead AI / ML Engineer	FreshSense Research Team	Approved	Sept 13, 2026
Product Design Director	UI/UX Systems Group	Approved	Sept 13, 2026
Software Architect	Full-Stack Infrastructure	Approved	Sept 13, 2026